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Current flow and energy balance during the evolution of instabilities in the plasma focus

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Abstract

The unique diagnostic system at the plasma focus PF-1000 at the IPPLM in Warsaw, operating at 2 MA and 10^{11} neutron yield, provides interferometry, x-ray, magnetic probes and neutron diagnostics together with registration of the voltage, current and its time derivative. With these diagnostic data we determined the boundaries of the dense plasma column from the interferograms and calculated the inductance for a single shot. The value of the current, voltage, current derivative, inductance and its derivative enabled calculation of the energy delivered to the pinch and energy released by the change of the inductance. The energy released during the expansion of the constriction was compared with the energy needed for acceleration of the deuterons producing neutrons calculated from the neutron pulse registered during and after the constriction expansion. The energy released by the expansion of the constriction is sufficient for deuteron acceleration.

Keywords: plasma focus, plasma diagnostics, current distribution

(Some figures may appear in color only in the online journal)

1. Introduction

Z-pinches and plasma foci are efficient devices for realizing deuterium–deuterium reactions and producing fusion neutrons [1, 2]. The mechanisms of the acceleration of the deuterons producing neutrons are still a matter of discussion as are many internal processes [2].

Interferometry is one of the methods for observing the structure and density of the plasma. It was used to measure plasma with density above 10^{23} m^{-3} in different plasma focus devices [1, 3].

Along with interferometry, magnetic probes are used. They register the temporal derivative of the azimuthal magnetic field at the anode front, which allows determination of the current density. The employment of the magnetic probes at the PF1000 facility was published in [4], stating that after the maximal pinch the mean current should flow along the skin layer of the plasma column.

The description of transportation of the energy to the pinched plasma is discussed in [1, 2, 5, 6]. The transport is influenced by changes of the inductance. The inductance depends on the distribution of the current along the dense plasma column and according to Krauz *et al* [4] it changes with the length and radius of the dense plasma column.

The aim of this paper is to use the border of the dense plasma column observed in the interferometric pictures to estimate the possible energy delivered to and absorbed by the pinch in the time and region of the constriction, with the assumption of the current flowing along the skin layer of the dense plasma column.

2. Experimental setup

The Mather-type plasma focus PF-1000 at the IPPLM in Warsaw, operating at 2 MA with 10^{11} neutron yield, was investigated with interferometry, x-ray and neutron diagnostics and magnetic probes. Also current, its derivative and voltage signals were registered at the collector inside the device in front of the electrodes. The device and the obtained diagnostic results are described in [7].

For temporal resolved hard x-ray and neutron diagnostics, scintillators BC408 and photomultipliers Hamatsu are used,

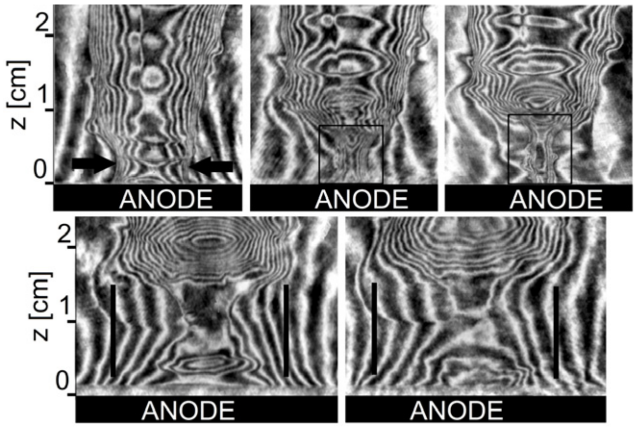

Figure 1. Interferograms of the investigated region of shot no. 9209. From the top left: (a) $t_1 = 42$ ns with the marked direction of the constriction; (b) $t_2 = 62$ ns with the marked region of the constriction; (c) $t_3 = 72$ ns minimal diameter with the marked region of the constriction; (d) $t_4 = 92$ ns expansion occurring with the marked assumed position of the flow of the mean of the current; and (e) $t_5 = 102$ ns marked assumed position of the flow of the mean of the current. The times are relative to the time of the maximal dip of the current derivative.

positioned in the radial (7 m) and upstream direction (7–84 m). Counters with activated silver are used for total neutron yield estimation.

A local interferometry setting produces 15 interferometric pictures with millimeter spatial resolution, representing the distribution of the line plasma density [8]. Their time step is 10–20 ns during a 210 ns period. A Nd:YLF laser with a pulse duration below 1 ns was used. With the interferograms, we can see the time evolution of the pinch dynamics and correlate it with the neutron and x-ray emission waveforms [9, 10].

During the experiments there was a hole in the anode center 5 cm in diameter, the gas filled was deuterium, the gas pressure was 180–200 Pa, the voltage was 24 kV and the mean neutron yield was of the order of 10^{10} [11].

3. Estimation of the energy balance during the constriction

We have investigated shot no. 9209 in the time and region of constriction evolution. The neutron yield in the pulse during the time of the constriction was 2×10^{10}. In the observed time span of 40–130 ns interferograms were registered in times 42, 62, 72, 92 and 102 ns and are shown in figure 1. The times of the interferograms are relative to the time of the maximal dip of the current derivative.

To estimate the energy balance of the entire pinch a formula derived from the electrical scheme of the pinch is used [5]:

$$UIt - I\frac{\mathrm{d}I}{\mathrm{d}t}L_0 t - I\frac{\mathrm{d}I}{\mathrm{d}t}L_\mathrm{p} t = I^2\frac{\mathrm{d}L_\mathrm{p}}{\mathrm{d}t}t + RI^2 t \quad (\mathrm{J}), \qquad (1)$$

where U is voltage, I is current, dI is its derivative, L_0 is the inductance of the electrode system (32 nH), L_p is the inductance of the plasma column, R is its resistance and t is the time duration of the pinch.

The resistance of the pinch and thus the energy losses in the resistance for temperatures over 100 eV can be neglected due to Spitzer's relationship.

For the time and spatial region of the constriction expansion, according to Inzhennik and Bobrova [5], the first and second term of (1) are constant due to the minimal change of current in the short observed time span Δt:

$$E_n = \left(-I\frac{\mathrm{d}I}{\mathrm{d}t}L_\mathrm{const}\Delta t\right) - I^2\frac{\mathrm{d}L_\mathrm{const}}{\mathrm{d}t}\Delta t \quad (J).$$
$$E_n = E_\mathrm{mag} - E_\mathrm{ind} \quad (J). \qquad (2)$$

The first term is designated as E_mag and represents the energy gain from the changes of the current in the inductance and the second term as E_ind which represents the changes of the energy caused by the changes of the inductance i.e. the current flow diameter. The sum of these two terms gives us the energy E_n released during the constriction expansion. L_const stands for the inductance of the plasma column in the constriction region.

We make an assumption that the current flows along a thin skin layer. During the decay of the constriction and production of the x-ray and neutron pulses (90th ns), the current flow rapidly expands to the distance of 1.5 cm from the anode axis. This model was presented by Trubnikov [12].

With this assumption, we calculate the time-resolved inductance L_p of the plasma column up to the distance of 1.3 cm from the anode front. In this region the constriction occurs. For the calculation, we use the formula for coaxial cylinder

$$\sum_1^n L_n = \sum_1^n \left(l_n \frac{\mu}{2\pi}\ln\frac{r_\mathrm{o}}{r_n}\right) \quad (H). \qquad (3)$$

With the number of the pixel columns in the actual picture n, l_n is the width of the pixel column in meters, μ is the magnetic constant ($4\pi \times 10^{-7}$ H m^{-1}), r is the inner radius determined by the borders of the plasma as seen in the interferograms and r_o is the radius of the outer electrodes (15 cm).

The inductance is calculated for each column of pixels and is then added up to get the inductance of the whole constriction region. By using this process for every interferogram and interpolating the data, we get time-resolved inductance.

The uncertainty of this calculation is given mainly by the uncertainty of tracing the plasma border in the interferometric picture. This uncertainty is estimated to be 20% and neglects the uncertainties of the other measured elements.

4. Results

The calculated time-resolved inductance of the constriction region L_const is shown in figure 2. It rises to 10 nH at the time of the maximal constriction and then rapidly decreases to 6 nH after the expansion. The mean velocity of the constriction implosion was estimated to be 2.3×10^5 m s^{-1} and the mean velocity of the explosion to be 3.8×10^5 m s^{-1}. Both were estimated from the interferograms as typical velocities for this device [9] with an uncertainty of 20% caused by the

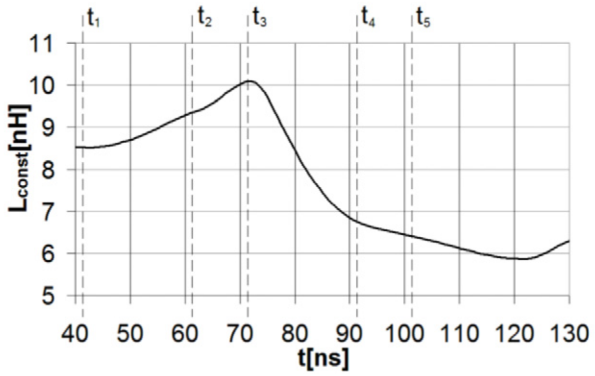

Figure 2. Inductance of the plasma column in the region of the constriction L_{const} in shot no. 9209 with marked times of the interferograms. The times are relative to the time of the maximal dip of the current derivative.

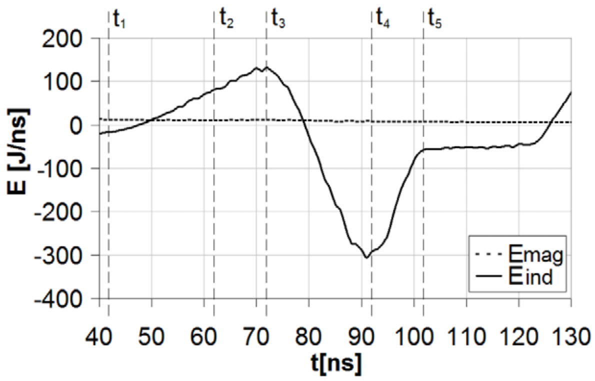

Figure 3. Time-resolved energies E_{mag} and E_{ind}, shot no. 9209 with marked times of the interferograms. The times are relative to the time of the maximal dip of the current derivative.

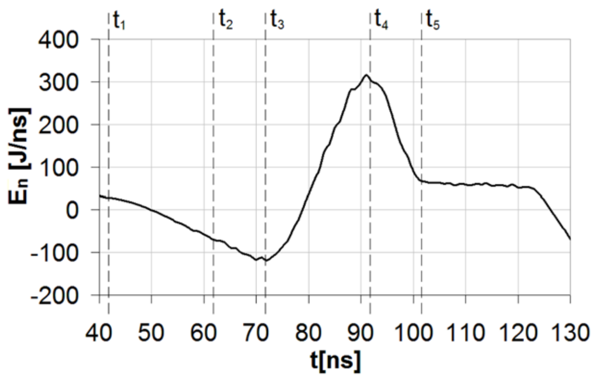

Figure 4. Time-resolved energy $E_n = E_{\text{mag}} - E_{\text{ind}}$, shot no. 9209 with marked times of the interferograms. The times are relative to the time of the maximal dip of the current derivative.

interpretation of the interferograms. The observed time span of the constriction expansion is 72–102 ns.

The calculated energies are shown in figure 3. The total energy E_n, given during the constriction expansion by the sum of E_{mag} and E_{ind}, is 4.5 kJ with 20% uncertainty and its time resolution is shown in figure 4.

The calculation of the energy E_n can be compared with the total energy of the deuterons in shot no. 9209. This energy is estimated using the formula for the neutron yield of the beam–target fusion mechanism

$$\mathrm{NY} = l\mu\rho n_{\mathrm{d}}, \tag{4}$$

where NY stands for neutron yield (2×10^{10}), l is the characteristic dimension of the target, μ is the density of the target, ρ is the cross-section of the reaction and n_{d} is the number of particles (deuterons) in the target. The target is the plasmoid itself, with parameters known from interferometry ($l = 0.02$ m and $\mu = 2 \times 10^{24}$ m^{-3}). The cross-section depends on the mean energy of the deuterons, which is estimated from the velocity of the neutrons, which is calculated from the shift of the neutron and x-ray pulses in the up-stream and side-on direction using a time-of-flight method [13].

With the number of deuterons $n_{\mathrm{d}} = 3 \times 10^{17}$ and their mean energy of 100 keV, we obtain the total energy of the deuterons to be 4.9 kJ with an uncertainty of 20%.

The total energy of the energetic electrons was considered to be negligible, due to their small number and mass compared to the deuterons.

The exact transportation method of the energy released by expansion of the constriction to the deuterons is unknown. For example, according to Aparicio *et al* [14], the energy E_n could cause turbulences leading to anomalous resistivity, which enables fast reconnection and acceleration of the deuterons.

5. Conclusion

The calculation shows that the energy released by the rapid expansion of the constriction region may be, within uncertainties, sufficient to accelerate the deuterons to velocities needed for producing fusion neutrons registered during the time of the constriction expansion.

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